

Quantum Programming: Getting Ready for the Real World

Explore the past, present, and future of open source quantum programming. Get a glimpse of the challenges it faces and the role of open source in its development.



The word ‘quantum’ has been largely confined to physics textbooks and discussions on superposition and entanglement in the past decades. However, it has now moved off the theoretical page into a world of computation where it promises to impact everything. Quantum programming, the act of creating code for quantum computers, is quickly emerging as the next significant technological frontier.

The rise of quantum programming

Classical computing, no matter how powerful, faces challenges in areas where problems are highly complex,

like simulations of molecules, global supply chain optimisation, or breaking modern encryption. Moore’s Law cannot work indefinitely. Quantum computing provides us with an entirely new path.

Quantum computers use qubits that can exist in many states simultaneously. Thus, rather than working through possible options one at a time, quantum computers can evaluate many possibilities at once. For tasks like factorisation, machine learning, and chemistry simulations, quantum algorithms promise exponential or nearly exponential speeds.

This has great relevance in the real world.

Healthcare and pharma: Quantum algorithms can speed up drug discovery by simulating molecular interactions at a level of precision far beyond what classical computers can achieve.

Finance: They can optimise portfolios or risk models involving an unfathomable number of variables.

Logistics and supply chains: Companies such as DHL and Airbus are currently investigating quantum tools for cost minimisation and efficiency.

Cybersecurity: While quantum programming may have implications for current encryption standards, it also enables the implementation of quantum-safe cryptography.